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I&C at Steam Turbine

ST Start-up Program

MAY01EC001, MAY20EC001

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Tasks

- Automatic steam turbine start with the help of a step program
- For ST start-up one manual release is always necessary (steam purity).
  A further manual releases in accordance with customer requirements (for example nominal speed) is often used.
- Time optimised (so fast as possible) and safe ST start under consideration of all limits and restrictions
- Covers the entire start-up procedures, this includes start of turning device operation up to initial loading
- Coordinates the entire I&C equipment for steam turbine start-up and shutdown

Function

- Inputs from other automation devices like:
  ST Stress function (DDT, X-Criteria, Z-Criteria),
  ST Controller, ST protection, Unit coordination, ..
- Activates the ST Controller
- SGC Steam Turbine Auxiliary System MAY20EC001
- SGC Steam Turbine (MW) MAY10EC001

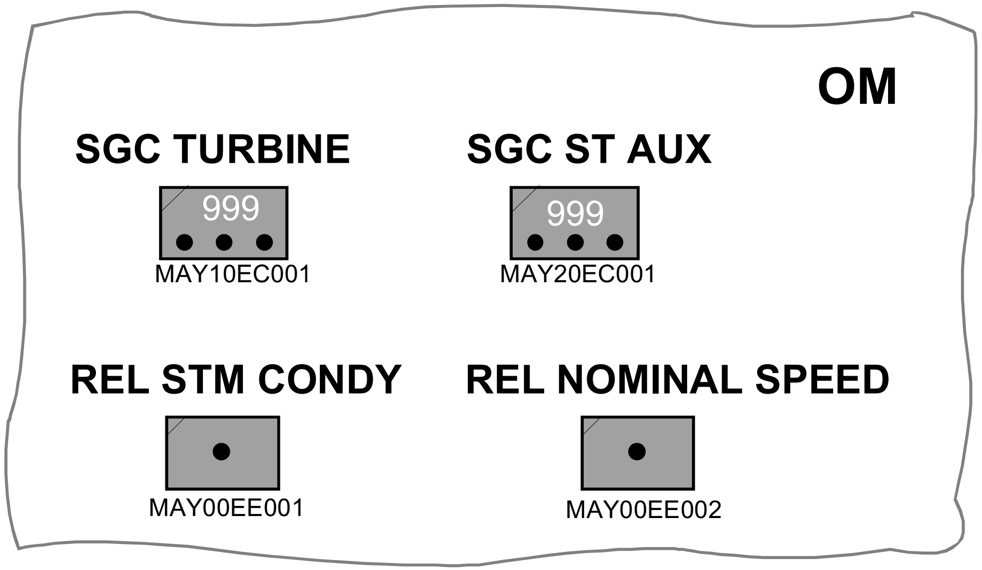

Fig. 1.1 Task and Function of Turbine Master Program

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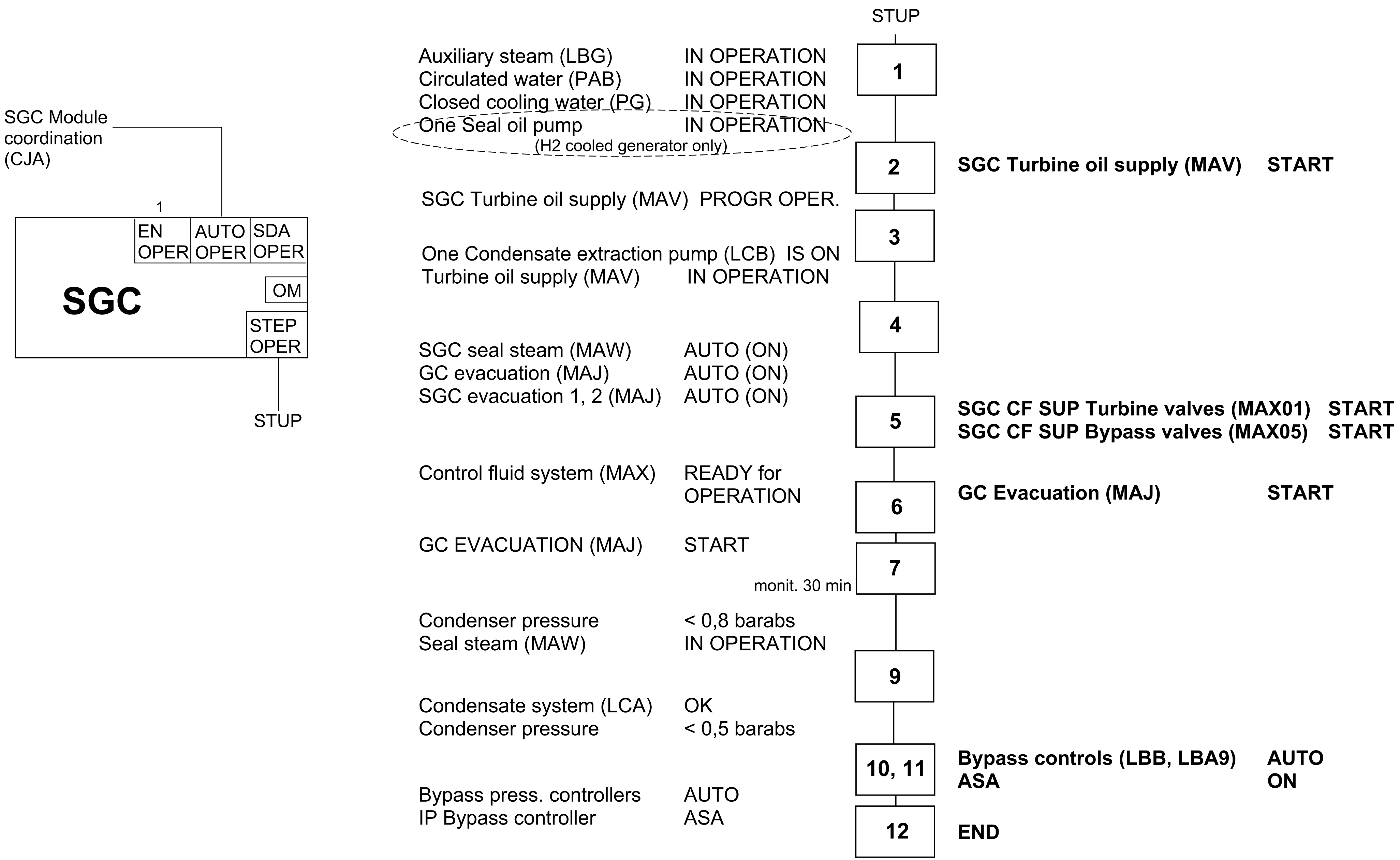

Fig. 2.1a SGC Steam Turbine (AUX) MAY20EC001 (simplified)

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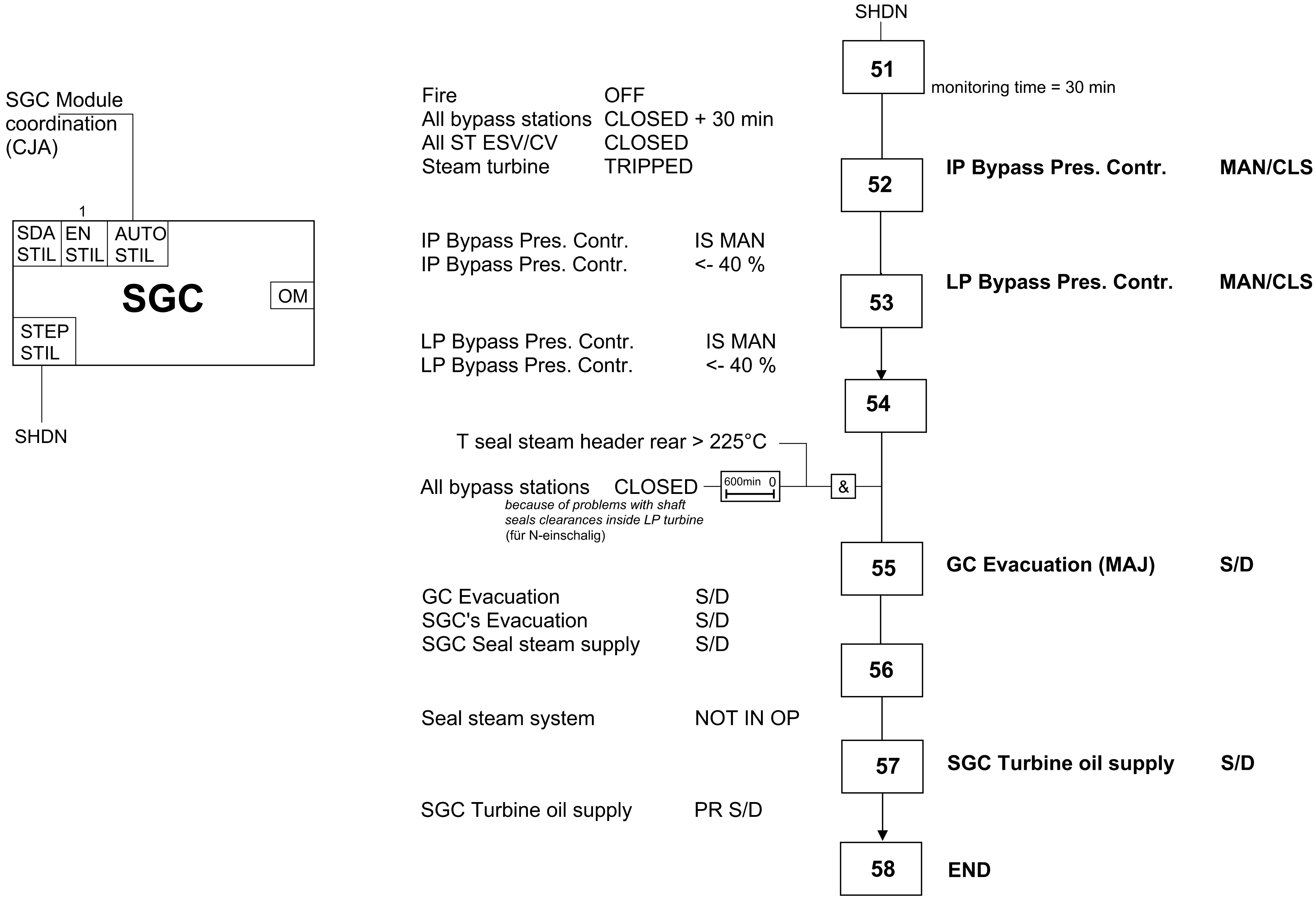

Fig. 2.1b SGC Steam Turbine (AUX) MAY20EC001 (simplified)

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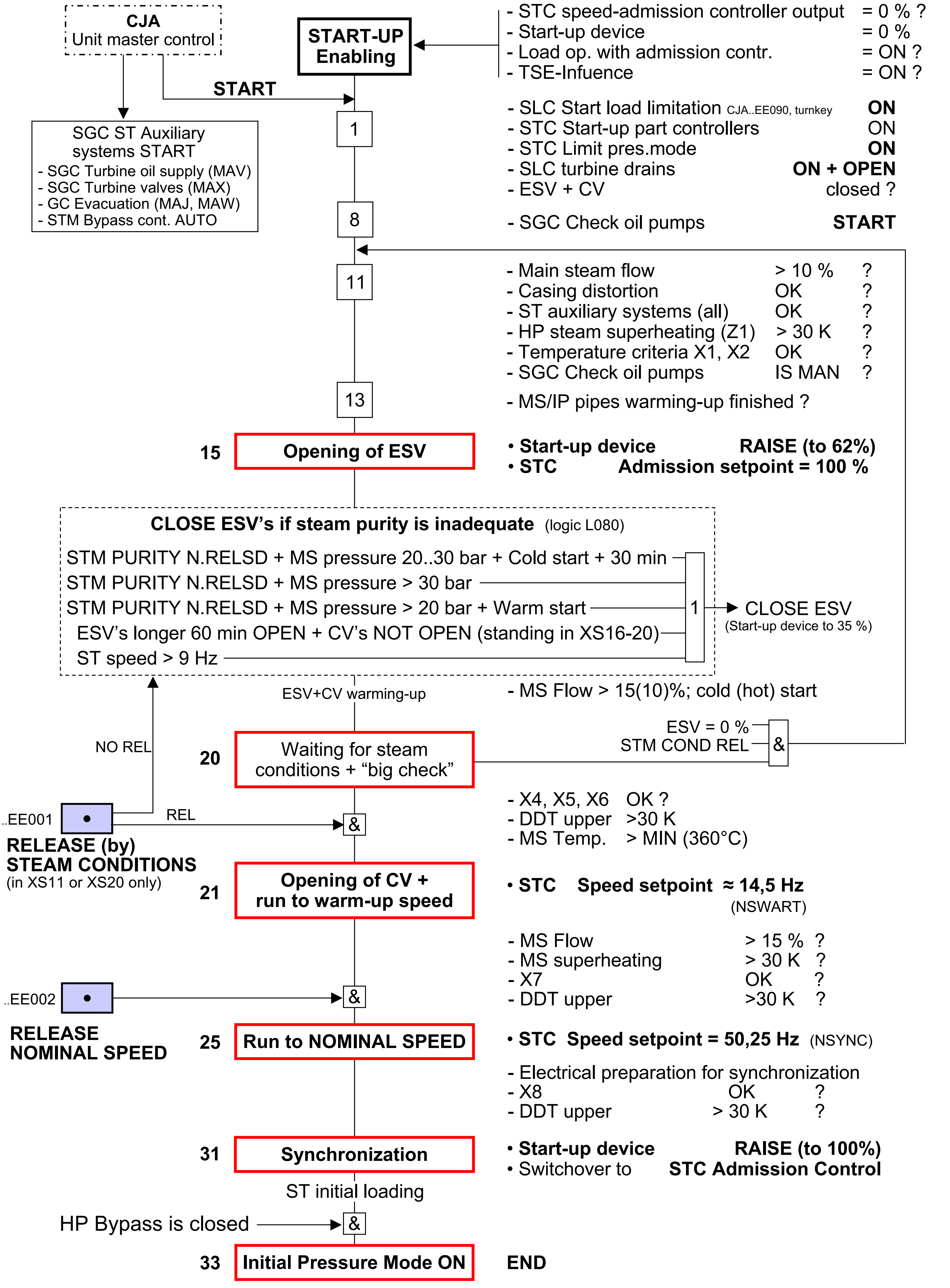

SGC Steam Turbine Start-up Program
MAY10EC001 (very simplified)

Fig. 3.11_KN

Table 1 Normal limit values and reasonably achievable values in normal operation for condensed steam sample

Parameter	Units	Normal limit[1)]	Reasonably Achievable Value [2)] (for condensing power plants)
Conductivity at 25 °C downstream of strongly acidic sampling cation exchanger, continuous measurement at sampling point	µS/cm	< 0.2	0.1
Sodium [3)] (Na)	µg/kg *	< 5	2
Silica (SiO_2)	µg/kg	< 10	5
Total iron (Fe)	µg/kg	< 20	5
Total copper[4)] (Cu)	µg/kg	< 2	1

1) To avoid any corrosion or loss of efficiency, it is advisable to keep the actual values below the Normal Limit values, preferably within the range of the Reasonably Achievable Values for normal operation.

2) The values in Reasonably Achievable Values are only achievable in continuous operation. In all other cases, the respective normal limit value is regarded as a maximum value for normal operation.

3) If solid alkalizing agents (NaOH, Na_3PO_4) are used only in case of abnormal operating conditions, continuous monitoring of sodium is not imperative. (See discussion of sodium).

4) No copper monitoring is necessary if the steam-water cycle is free of copper alloys (see discussion of Total Copper).

* 1 µg/kg = 1 part per billion (ppb)

Normal Operation

Recommended limits for impurities commonly found in turbine steam are given in Table 1. The normal limit values represent Siemens recommendations for reliable turbine operation.
These values are limits where experience has shown minimal deposition of salts in the dry regions of the turbine.
The reasonably achievable values provide extra assurance that corrosive impurities will not deposit and every reasonable effort should be made to operate at these values. The recommendations apply whenever steam is admitted to the turbine.
Low load does not protect the turbine from deposition.

The parameters given in Table 1 are for the impurities commonly found in steam power systems. If other impurities (not including the pH control agent and oxygen during oxygenated water treatment) are known to be present above 5 µg/kg (ppb), please consult Siemens for guidance on setting limits for those impurities.

Action Levels

It is not always possible to meet the normal values at justifiable expense, particularly during start-up of a plant. Therefore, action levels have been set up to rate the urgency of action to correct deviations from normal operating conditions.
The action levels represent undesirable conditions that should be corrected to normal within the time periods

Action level 1	cumulative time per year < 2000 h
Action level 2	cumulative time per year < 500 h
Action level 3	cumulative time per year < 80 h
Action level 4	cumulative time per year < 0 h

For further details: see steam turbine manual!

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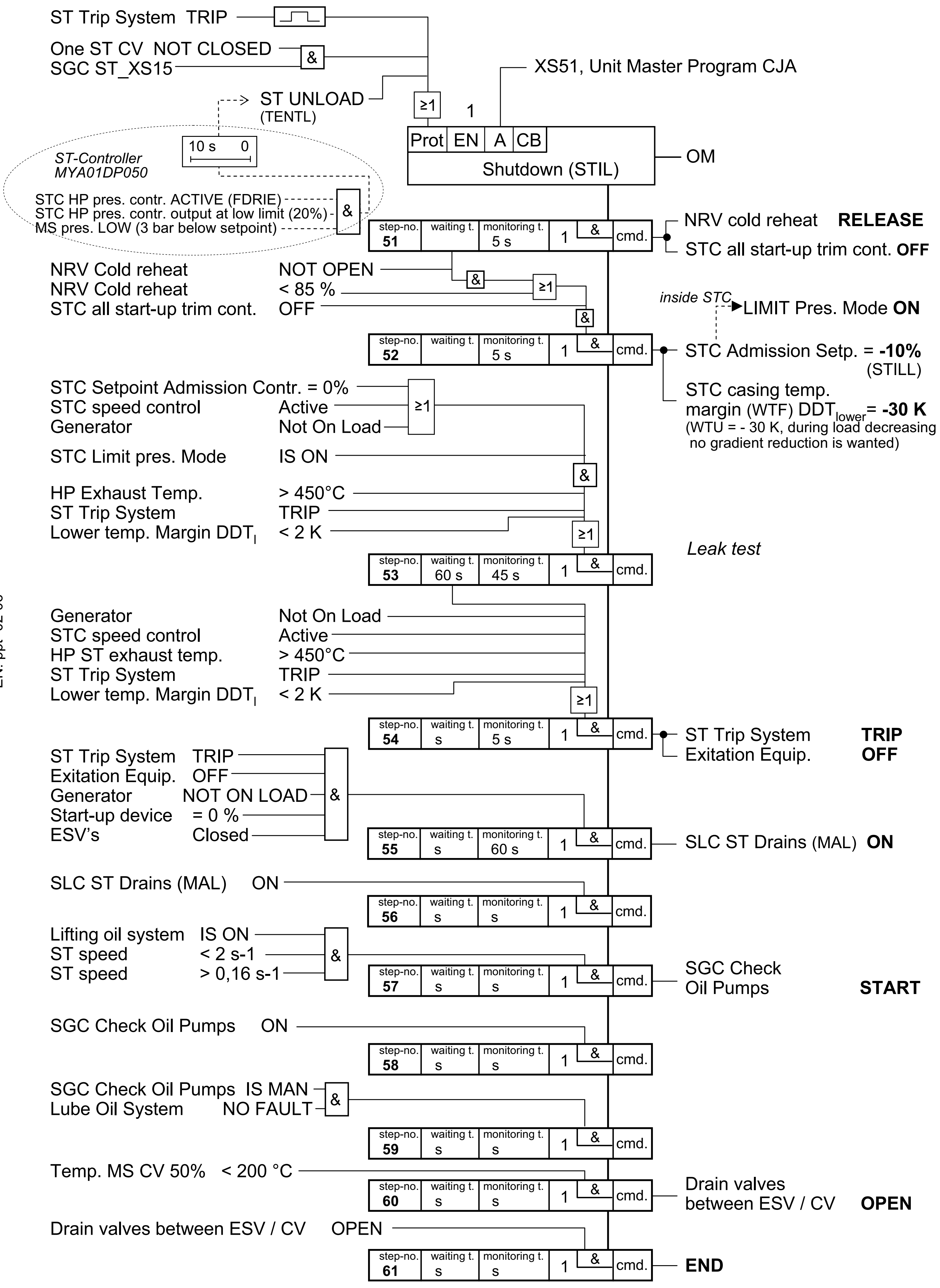

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SGC ST MAY10EC001, Shutdown Programm Fig. 5.1_KN

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10MKA	ST Generator
11MKA	GT Generator GT
A OPER	SGC ON and automatic start operation program
A_OFF	Automatic OFF command for SGC
ACCL	Acceleration
ADM	Admission
AHD	Ahead
ALLESV_OP	All emergency stop valves open
ALLESV_CLSD	All emergency stop valves are closed
ASA	Automatic Setpoint Adjuster
ATT	Automatic Turbine Tester
AUT PARAL-LELING DEV	Automatic synchronisation device
BB	Black box
B/P	Bypass
BEF	Before
BEH	Behind
BLD	Blading
BP	Bypass
BRG	Bearing
BRG ES	Bearing exciter side
BRG TS	Bearing turbine side
BTM	Bottom
BUT-V	Butterfly valve
BYP	Bypass
CB	Circuit breaker
CB	Check back
CC	Combined cycle
CCPP	Combined Cycle Power Plant
CCW	Closed cooling water
CCWP	Closed cooling water pump
CCWS	Closed cooling water system
CEP	Condensate extraction pump
CF	Control fluid
CF SUP	Control fluid supply
CIRC WTR	Circuit water
CKVS TURB EXTR	Check valves turbine extraction
CLG	Cooling
CLSD	Closed
CLSESV	Close command for emergency stop valves
CMD	Command
CNT-V	Control valve
CONTRL	Control
CRH	Cold reheat
CRH NRV	Cold reheat non return valve
CSG	Casing
CSG F TOP/BOT	Casing front side top/bottom
CSG INL LFT	Casing inlet left
CSG R TOP/BOT	Casing rear side top/bottom
CTRL	Control
CTRL-A	Control automatic
CTRL-V	Control valve
CV	Control valve
CWS	Cooling water system
D/STR	Downstream
D-V	Drain valve
DDT	Temperature margin
DEV	Device
DIFF	Difference
DIS	Discharge
DP	Delta p (pressure difference)
DRN	Drain
DRN_OK	Drains OK
DRN-V	Drain valve
DT	Delta Temperature (=ΔT)
DYD	Delayed
EHA	Electro-hydraulic Actuator
ENAB SPC	Enabling setpoint control
ES-DAMP	Emergency stop butterfly valve
E-SHAFT	E Turbine shaft, LP part
ESV	Emergency stop valve
ESV_OP	Emergency stop valves open
ESVS	Emergency stop valves
EXC	Excitation
EXTN	Extraction
FCV	Main steam control valve
FIL-V	Filter valve
FLTD	Faulted
FLT	Fault
FR HP-CV	Front high pressure control valve
GC	Group control
GEN	Generator
GEN CB	Generator circuit breaker
GEN_OK	Generator OK
GLND	Gland
GOV	Governor
GT	Gas Turbine
GTR.CKT.BRK.	Generator transformer circuit breaker
HDU	HP Bypass Station
HP	High Pressure
HP B/P	High Pressure Bypass
HP RED STAT	High Pressure Bypass Station
HPS	HP shaft
HRH	Hot reheat
HRSG	Heat recovery steam generator
IN OPER	In operation
INJ	Injection
IP	Intermediate pressure
IPS	IP shaft
L	Level
LFT	Left
LIM_OPN	Limiter at open
LIMIT ON	Limitation (for lift) effective (reached)
LIQ	Liquid
LMT	Limit
LP	Low pressure
LVL	Level
LWR	Lower
MARG	Margin
MEAS	Measurement
MID	Middle
MS	Main Steam
N FLTD	Not faulted
NOT SEL	Not selected
NRV	Non return valve
NR-V	Non return valve
O/S	Overspeed
O/S PROT	Overspeed protection
OM	Operating and monitoring system
OPN	Operation
OS	Admission Setpoint
P	Pressure
P-V	Pilot valve / pneumatic valve
PB	Push button
PL coil	Pilot coil
PMP	Pump
POS	Position
POSN	Position
PPS	Pumps
PR	Pressure
PR	Program
PRES	Pressure
PROP	Program operation
PROT	Protection
PROTN	Protection
RDY F OP	Ready for operation
READY FO	Ready for operation
REHT	Reheat
REL	Relative
REL_SU	Release shut down
RELS	Release
RHT	Reheat
S/D	Shut down
S/HTG	Superheating
S/O	Shut off valve
S/UP	Start-up
SEL	Selection
SERVWTR	Service cooling water
SETPOINT-C	Setpoint control
SLC	Sub loop control
SOV	Solenoid valve
SP	Setpoint
SPEED ACCL	Speed acceleration
SS	Steam Turbine Trip
ST	Steam Turbine
ST CONDY	Steam conditions
ST OP	Steam turbine operation
ST UNIT CB	Station unit circuit breaker
STC	Steam Turbine Controller
STILL_F	Shut down criteria for SGC ST
STILL_G	Shut down criteria for SGC ST
STM	Steam
STM CONDY	Steam conditions
SUP	Supply
SYN-SLCTR	Synchronising selector
T	Temperature
T SAT	Saturated steam temperature
TEA3	Turbine extraction A3
TEMP MARG	Temperature margin from TSE function
TM GROOVE HPS	Temperature middle groove (= mean temperature) High pressure shaft
Tm-gr	Temperature middle groove (= mean temperature)
TPA	Automatic Turbine Tester (= ATT)
TS	Turbine side
TSE	Turbine Stress Evaluator
TURB/BYP	Steam turbine bypass
U/D	Up/down
U/S	Upstream
U/STRM	Upstream
UNSEL	Un-selected, =not selected
UPR	Upper
US	Upstream
VERKR	Casing distortion (Verkrümmung)
VERKR_OK	Casing distortion OK
VIP	Vibration
VLV	Valve
VOLT	Voltage
W/U	Warm-up
WTR	Water
WTR INJ CV	Water injection control valve

Fig. 6.1 Abbreviations used inside SGC ST

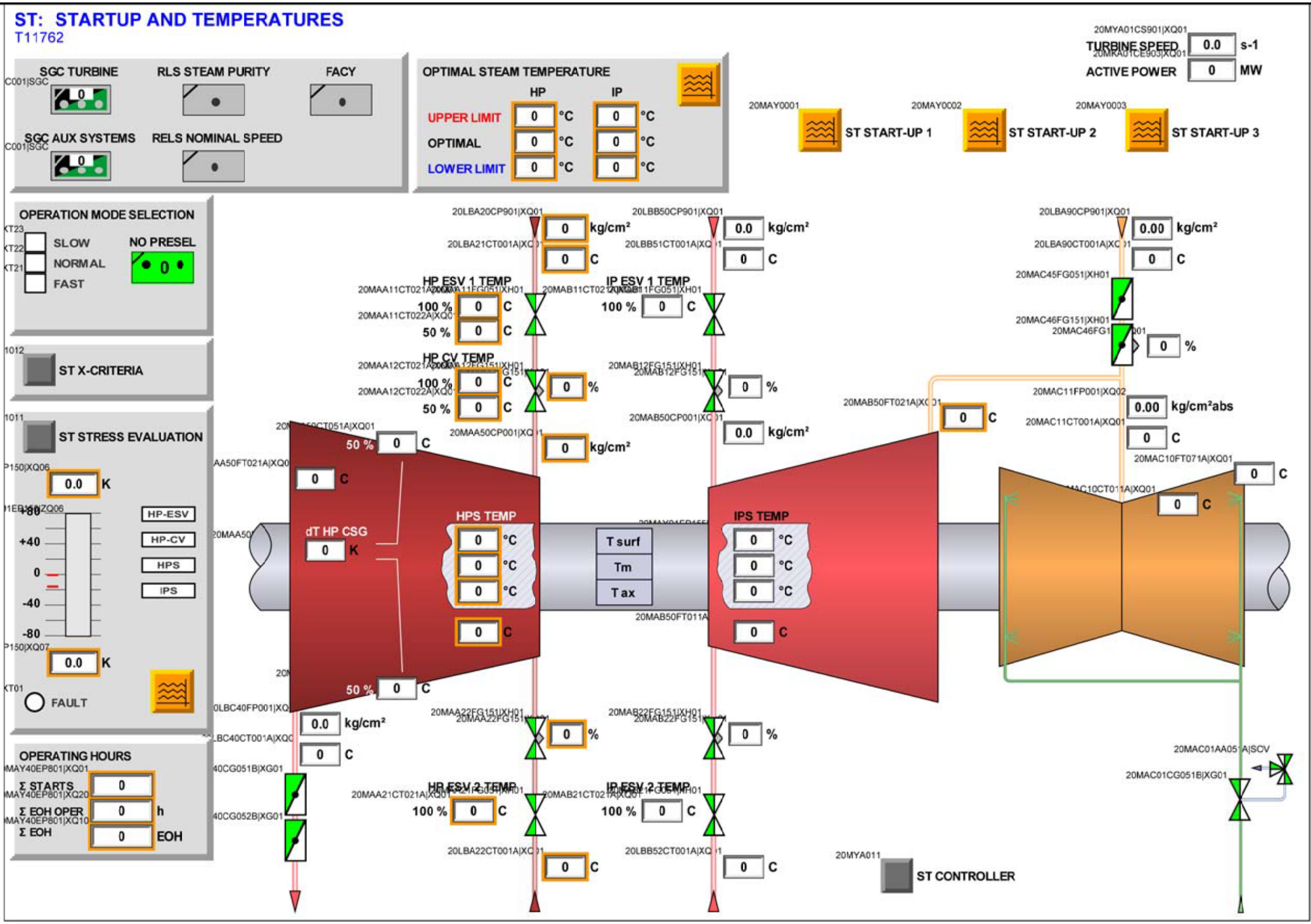

Fig. 7.1

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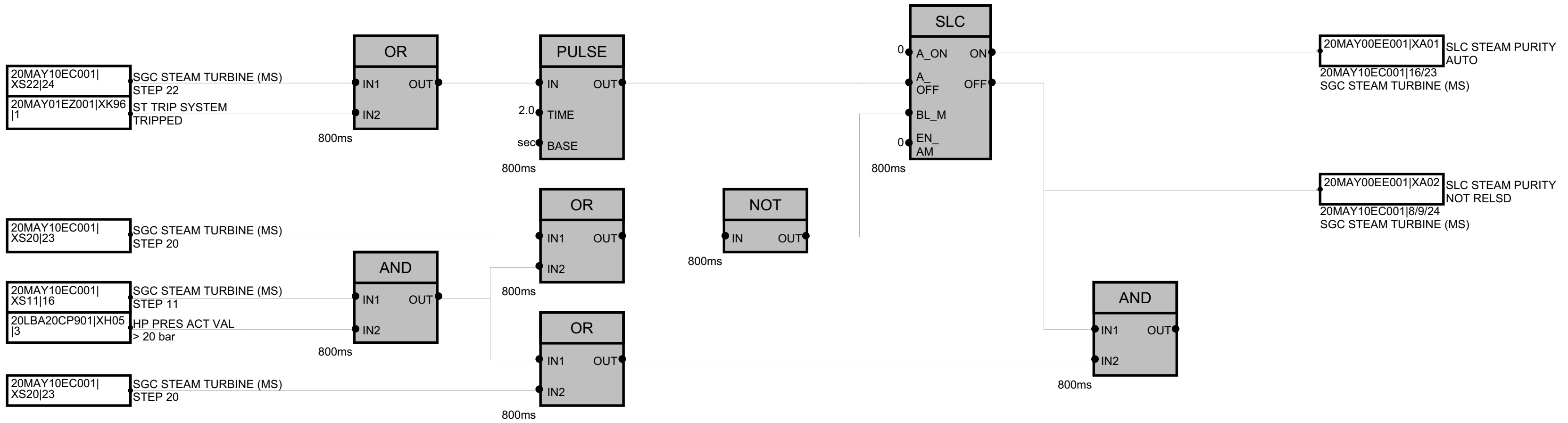
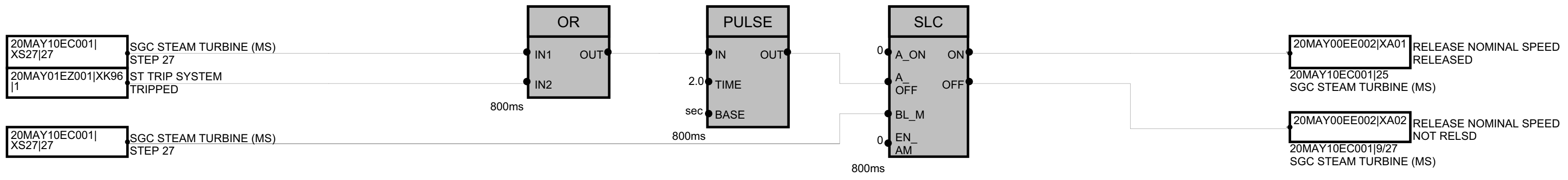
				Drawn By							+
				Checked By	Zeiß	INCHEON 2 CCPP		Plant Display			Page 1
State	Change	Date	Editor	Standard	PG P11M3/L47	2008-02-01_MAZ01-ZCRU01_A					Pg 1



				Date	08/01/01	KOMIPO Co., Ltd.	SLC STEAM PURITY		EFA00	= 20MAY00EE001
				Drawn By	Karg					+ S1_AP2
				Checked By	Zeiß	INCHEON 2 CCPP	Area Level Diagram	XMAY81		Page 1
State	Change	Date	Editor	Standard	PG P11M3/L47	2007-08-01_MAY80-XMAY81_C				Pg 1

FUP 1



				Date	08/01/01	KOMIPO Co., Ltd.	RELEASE NOMINAL SPEED		EFA00	= 20MAY00EE002
				Drawn By	Karg					+ S1_AP2
				Checked By	Zeiß	INCHEON 2 CCPP	Area Level Diagram	XMAY81		Page 1
State	Change	Date	Editor	Standard	PG P11M3/L47	2007-08-01_MAY80-XMAY81_C				Pg 1

FUP 2